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α-Glucosidase inhibition by luteolin: Kinetics, interaction and molecular docking

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ABSTRACT

α-Glucosidase is a critical associated enzyme with type 2 diabetes mellitus in humans. Inhibition of α-glucosidase is important due to the potential effect of down regulating glucose absorption in patients. In this study, the inhibitory activity of flavone luteolin on α-glucosidase and their interaction mechanism were investigated by multispectroscopic methods along with molecular docking technique. It was found that luteolin reversibly inhibited α-glucosidase in a noncompetitive manner with an IC_{50} value of $(1.72 \pm 0.05) \times 10^{-4}$ mol L^{-1}, and the inhibition followed a multi-phase kinetic process with a first-order reaction. Luteolin had a strong ability to quench the intrinsic fluorescence of α-glucosidase through a static quenching procedure. The positive values of enthalpy and entropy change suggested that the binding of luteolin to α-glucosidase was driven mainly by hydrophobic interactions, and the binding distance was estimated to be 4.56 nm. Analysis of synchronous fluorescence, circular dichroism, and Fourier transform infrared spectra demonstrated that the binding of luteolin to α-glucosidase induced rearrangement and conformational changes of the enzyme. Moreover, the results obtained from molecular docking indicated that luteolin had a high affinity close to the active site pocket of α-glucosidase and indirectly inhibited the catalytic activity of the enzyme.

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1. Introduction

Diabetes mellitus, as a health-threatening concomitant disease throughout the world, is a chronic metabolic disease associated with disorders of carbohydrate metabolism and characterized by hyperglycemia [1–3]. One of the therapeutic approaches to reduce postprandial hyperglycemia is to retard digestion and absorption of dietary carbohydrates by inhibiting digesting enzymes, such as α-glucosidase and α-amylase, in the digestive organs [4]. α-Glucosidase hydrolyzes α-(1-4)-linked D-glucose residues from the non-reducing end of α-glucoside [5], which is the final step in the digestion of polysaccharides and disaccharides. α-Glucosidase activity is directly related to levels of plasma glucose, and inhibition of α-glucosidase is important due to the potential effect of down regulating glucose absorption in patients with type 2 diabetes mellitus. Although many α-glucosidase inhibitors have been reported, most of the α-glucosidase inhibitors are sugars or derivatives of sugar moieties such as acarbose which is the first member of α-glucosidase inhibitors approved for the treatment of type 2 diabetes. There are only few non-saccharide compounds including substrate analogs [6,7], alkaloids, phenolics, curcuminoids, terpinoids, and anthocyanins from natural sources [8], which are taken orally for the treatment of diabetes by inhibiting α-glucosidase activity.

The lack of the nature of the interaction between α-glucosidase and inhibitors has made it a difficult task to discover good lead compounds. Consequently, exploring the binding behavior and inhibitory effect of inhibitors with α-glucosidase are of great importance for drug–enzyme interactions and therapeutic applications [9]. It can provide useful information to clarify the binding mechanism of inhibitors to α-glucosidase at molecule level. In recent years, the interaction between flavonoids and serum albumins has attracted much attention [10–12]. However, the mechanism of interaction between inhibitors and α-glucosidase (as another important target protein) has not received much attention.

Flavonoids, a class of natural drugs with high biological activity, are abundant in various plants. They have been reported to have protective effects in development of diabetes [13,14] as well as a mitigation effect of diabetic complications [15,16]. Luteolin (3′,4′,5,7-tetrahydroxyflavone, structure shown in Fig. 1A), a flavonoid compound containing a structure of the C-5 and C-7 hydroxyl groups in A ring, is present in some vegetables, fruits and medicinal herbs such as celery, green pepper and carrot [17], and has been shown to exhibit antioxidant, anti-inflammatory, anti-allergy, and anti-cancer activities [18]. Additionally, luteolin has recently been reported to act as an inhibitor of α-glucosidase [19,20]. Tadera et al. [21] have compared the inhibitory activity of six groups of flavonoids against yeast α-glucosidases and

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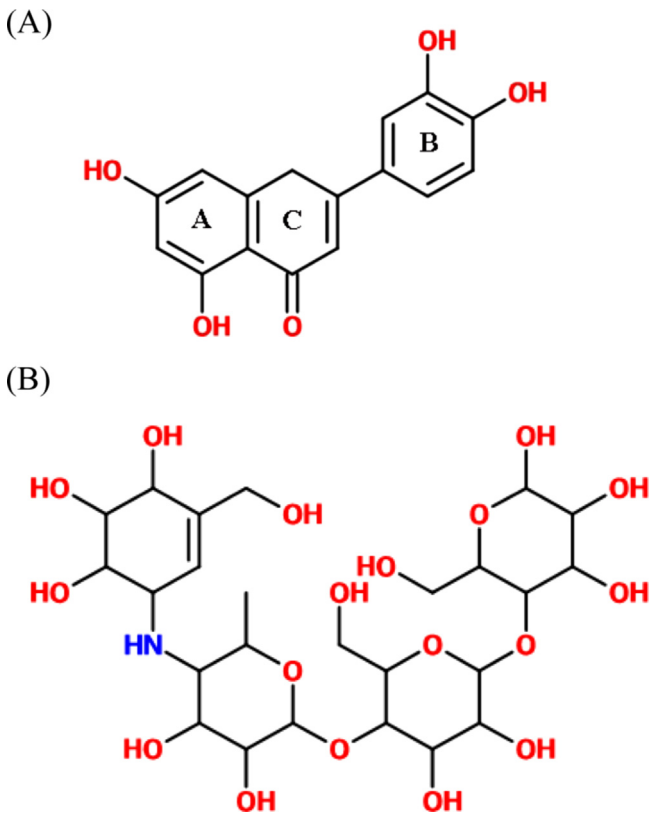


Fig. 1. Structures of luteolin (A) and acarbose (B).

determined the chemical structures of flavonoids responsible for the inhibitory activity. According to the tested results of Kim et al. have compared the α-glucosidase inhibiting activities of twenty-one naturally occurring flavonoids and found that luteolin was the strongest inhibitor [22]. Although there are some reports describing the α-glucosidase inhibitory properties of luteolin, most of these studies were limited to the enzymatic activity assay. The inhibitory mechanism of luteolin on α-glucosidase has not been reported.

Many spectroscopic methods including fluorescence, UV–vis absorption, circular dichroism (CD) and Fourier transform infrared (FT-IR) spectroscopy have been employed to investigate the drug–protein interactions and the conformational changes of proteins, because of their high sensitivity, reproducibility and convenience [23]. The aim of this study was to determine the inhibitory effects of luteolin on α-glucosidase and investigate their interaction. The inhibitory activity and type, binding mechanism, and thermodynamic parameters were characterized using UV–vis absorption, fluorescence and molecular modeling methods. The conformational changes of α-glucosidase induced by luteolin were analyzed by synchronous fluorescence, CD and FT-IR spectroscopic approaches. The results from this study have provided new insights into the inhibition of this important enzyme and suggested some significant clues to clinical applications for luteolin as an α-glucosidase inhibitor.

2. Materials and methods

2.1. Materials

α-Glucosidase (EC 3.2.1.20) from baker's yeast was obtained from Tokyo Chemical Industry Co. (Shanghai, China) and dissolved in 0.1 mol L^{-1} sodium phosphate buffer of pH 7.0. Acarbose and p-nitrophenyl-α-D-glucopyranoside (pNPG) were purchased from Sigma–Aldrich Co. (St. Louis, MO, USA), and their stock solutions (1.00×10^{-2} mol L^{-1}) were prepared with sodium phosphate buffer (pH 7.0). Luteolin (analytical grade) was obtained from Aladdin Chemical Co. (Shanghai, China), and its stock solution (1.00×10^{-2} mol L^{-1}) was made in ethanol. Although high concentrations of ethanol affect protein structure [24], the amount of ethanol used in the study had no inhibitory effect on α-glucosidase, and used in making the UV–vis absorption, fluorescence, CD, and FT-IR spectra measurements was very small (the maximum ethanol concentrations of sample solution in fluorescence, CD and FT-IR spectra measurements were 1.1%, 1.2% and 1.2%, respectively). All other reagents and solvents were of analytical reagent grade, and ultrapure water was used throughout the experiment. All stock solutions were stored at 0–4 °C.

2.2. Apparatus

The UV–vis absorption measurements were performed with a Shimadzu UV-2450 spectrophotometer (Shimadzu, Japan) using a 1.0 cm cell. The fluorescence spectra were measured on a Hitachi spectrofluorometer model F-7000 (Hitachi, Japan) equipped with a 150 W xenon lamp and a thermostat bath, using a 1.0 cm quartz cell. The CD spectra were recorded on a Bio-Logic MOS 450 CD spectrometer (Bio-Logic, Claix, France) using a 1.0 mm path length quartz cuvette. Infrared spectra were recorded with a FT-IR spectrometer (Thermo Nicolet-5700, USA) equipped with a germanium attenuated total reflection (ATR) accessory, a DTGS KBr detector and a KBr beam splitter. An electronic thermostat water bath (Shanghai Yuejin Medical Instrument Co., Shanghai, China) was used for controlling the temperature. All experiments, unless specified otherwise, were carried out at room temperature.

2.3. α-Glucosidase activity assay

The α-glucosidase activity was measured using a slightly modified method of Ohta et al. [25]. A series of assay solutions dissolved in 0.1 mol L^{-1} sodium phosphate buffer of pH 7.0 containing different concentrations of luteolin and fixed concentration of α-glucosidase (1.0×10^{-6} mol L^{-1}) were prepared and incubated for 3 h at 37 °C. The assay was initiated by adding the luteolin–enzyme complex solution to the substrate (pNPG, 5.0×10^{-4} mol L^{-1}), and the final volume of the reaction system was 2.0 mL. Subsequently, absorbance of the mixture at 405 nm was recorded every 30 s. The enzymatic activity assayed without luteolin was defined as 100%. The α-glucosidase inhibitory activity of luteolin was expressed as the median effective concentration for inhibitory activity (IC_{50}), that is, the amount of tested luteolin required for a 50% decrease in α-glucosidase inhibitory activity [26]. Relative enzymatic activity (%) = (slope of reaction kinetics equation obtained by reaction with inhibitor)/(slope of reaction kinetics equation obtained by reaction without inhibitor) × 100%.

2.4. Inhibitory kinetic analysis

To describe the noncompetitive inhibition mechanism, the Lineweaver–Burk equation in double reciprocal form was expressed as follows:

$$\frac{1}{\nu} = \frac{K_m}{\nu_{max}}\left[1 + \frac{[I]}{K_i}\right]\frac{1}{[S]} + \frac{1}{\nu_{max}}\left[1 + \frac{[I]}{\alpha K_i}\right] \quad (1)$$

and secondary plot was constructed as:

$$\text{Slope} = \frac{K_m}{V_{max}} + \frac{K_m[I]}{V_{max}K_i} \quad (2)$$

where ν is the enzyme reaction rate in the absence and presence of luteolin. K_i and K_m are the inhibition constant and

Michaelis–Menten constant, respectively. α is the ratio of the uncompetitive inhibition constant to competitive inhibition constant, and the value of α is 1 as an noncompetitive inhibition. [*I*] and [*S*] are the concentrations of inhibitor and substrate, respectively. The secondary plots of slope vs. [*I*] are linearly fitted, assuming a single inhibition site [27].

2.5. Fluorescence spectra measurements

The fluorescence measurements were performed at different temperatures (298, 304, and 310 K) in the range of 300–500 nm with an excitation wavelength at 280 nm. The excitation and emission bandwidths were both set at 2.5 nm. α-Glucosidase concentration was kept at 2.0×10^{-6} mol L^{-1}, and luteolin concentrations were varied from 0 to 3.54×10^{-4} mol L^{-1}. Appropriate blanks corresponding to the buffer were subtracted to correct background fluorescence.

The synchronous fluorescence spectra were obtained by setting the excitation and emission wavelength interval ($\Delta\lambda$) at 15 and 60 nm, at which the spectrum only showed the spectroscopic behavior of tyrosine and tryptophan residues of α-glucosidase, respectively.

In the present study, all fluorescence intensities were corrected for the absorption of excited light and the re-absorption of emitted light. The following relationship was used to correct the inner-filter effect [28]:

$$F_c = F_m e^{(A_1+A_2)/2} \tag{3}$$

where F_c and F_m are the corrected and measured fluorescence, respectively. A_1 and A_2 are the absorbance of luteolin at excitation and emission wavelengths, respectively. The intensity of fluorescence used in this work was the corrected fluorescence intensity.

To elaborate the probable quenching mechanism between luteolin and α-glucosidase, the fluorescence quenching data were analyzed by the Stern–Volmer equation [9]:

$$\frac{F_0}{F} = 1 + K_{SV}[Q] = 1 + K_q\tau_0[Q] \tag{4}$$

where F_0 and F are the fluorescence intensities (344 nm) of α-glucosidase in the absence and presence of quencher, respectively. K_{SV} is the Stern–Volmer quenching constant ($K_{SV} = K_q\tau_0$), which was determined by linear regression of a plot of F_0/F against [Q]. K_q is the quenching rate constant of biomolecule, τ_0 is the average lifetime of the fluorophore without quencher, and the value of τ_0 of the biopolymer is 10^{-8} s [29], and [Q] is the concentration of luteolin.

Assumed small molecules bind independently to a set of equivalent sites on a macromolecule, the binding constant (K) and the number of bound luteolin to α-glucosidase (n) were determined by plotting the double logarithm regression curve using the following equation [30]:

$$\log \frac{F_0 - F}{F} = \log K + n \log [Q] \tag{5}$$

The thermodynamic parameters were calculated in order to further characterize the intermolecular forces between α-glucosidase and luteolin. If the temperature does not vary significantly, the enthalpy change (ΔH°) can be regarded as a constant, then its value and entropy change (ΔS°) value can be evaluated from the van't Hoff equation as follows:

$$\log K_a = -\frac{\Delta H^\circ}{2.303RT} + \frac{\Delta S^\circ}{2.303R} \tag{6}$$

where R is the gas constant, and the temperatures used were 298, 304 and 310 K. The value of free energy change (ΔG°) can be obtained from the following equation:

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ \tag{7}$$

2.6. Energy transfer

According to Förster's non-radiative energy transfer theory [31], the fluorescence emission spectrum of α-glucosidase and UV absorbance spectrum of luteolin were obtained under the condition of the same concentration, for all the samples. The overlapping area of the two spectra and the distance (r) between the donor and the acceptor were calculated. In this study, the concentrations of luteolin and α-glucosidase were fixed at 2.00×10^{-6} mol L^{-1}, the UV–vis absorption spectrum of luteolin and the emission spectrum of α-glucosidase were recorded in the range of 300–500 nm.

2.7. CD studies

CD measurements of α-glucosidase in the presence of luteolin were made at wavelengths between 200 and 250 nm under constant nitrogen flush. The CD spectra were measured by keeping the concentration of α-glucosidase constant (2.0×10^{-6} mol L^{-1}) while varying luteolin concentration from 0 to 4.0×10^{-4} mol L^{-1}. All observed CD spectra were recorded in pH 7.0 sodium phosphate buffer at room temperature and corrected for buffer signal, and results taken as CD ellipticity in mdeg. The contents of different secondary structures of α-glucosidase, e.g., α-helix, β-sheet, β-turn and random coil, were analyzed by the online SELCON3 program (http://dichroweb.cryst.bbk.ac.uk/html/home.shtml).

2.8. FT-IR measurements

The FT-IR spectra of α-glucosidase (2.0×10^{-6} mol L^{-1}) and its luteolin complex were measured in the range of 1800–1400 cm^{-1} in sodium phosphate buffer, pH 7.0 at room temperature. All spectra were taken via the ATR method with the resolution of 4 cm^{-1} and 60 scans. The final concentration of luteolin used was 4.0×10^{-4} mol L^{-1}. The corresponding absorbance contributions of free luteolin and buffer solutions were recorded and subtracted with the same instrumental parameters. The secondary structure compositions of α-glucosidase and its luteolin complex were estimated by the FT-IR spectra and curve-fitted results of amide I band. Data were analyzed using Omnic software (version 7.2, Nicolet Instrument Co, Madison, WI, USA), and the curves were fitted using Origin software (version 6.0, Origin Lab, Northampton, MA, USA).

2.9. Homology modeling and molecular docking studies

The X-ray crystal structures of a few bacterial α-glucosidase have been reported, however, the 3D structure of α-glucosidase used in biological assays from yeast has never been reported [32]. To find a proper structural template for homology modeling, we searched for the Protein Data Bank (PDB) from the protein sequence data bank (http://www.ncbi.nlm.nih.gov/protein/), and used BLAST algorithms with the amino acid sequence of the target as input. The 3D structure of luteolin and acarbose was generated in Sybyl × 1.1 (Tripos Inc., St. Louis, USA), and the homology model was used as the receptor model in the virtual screening with docking simulations. All the water molecules were removed, whereas Gasteiger charges and essential hydrogen atoms were added by using the AutoDock tools (version 4.2) [33]. The rotatable bonds in the ligand were assigned with AutoDock Tools, and the ligand docking was performed with the AutoDock Lamarckian Genetic Algorithm (LGA, runs 100). The homology model

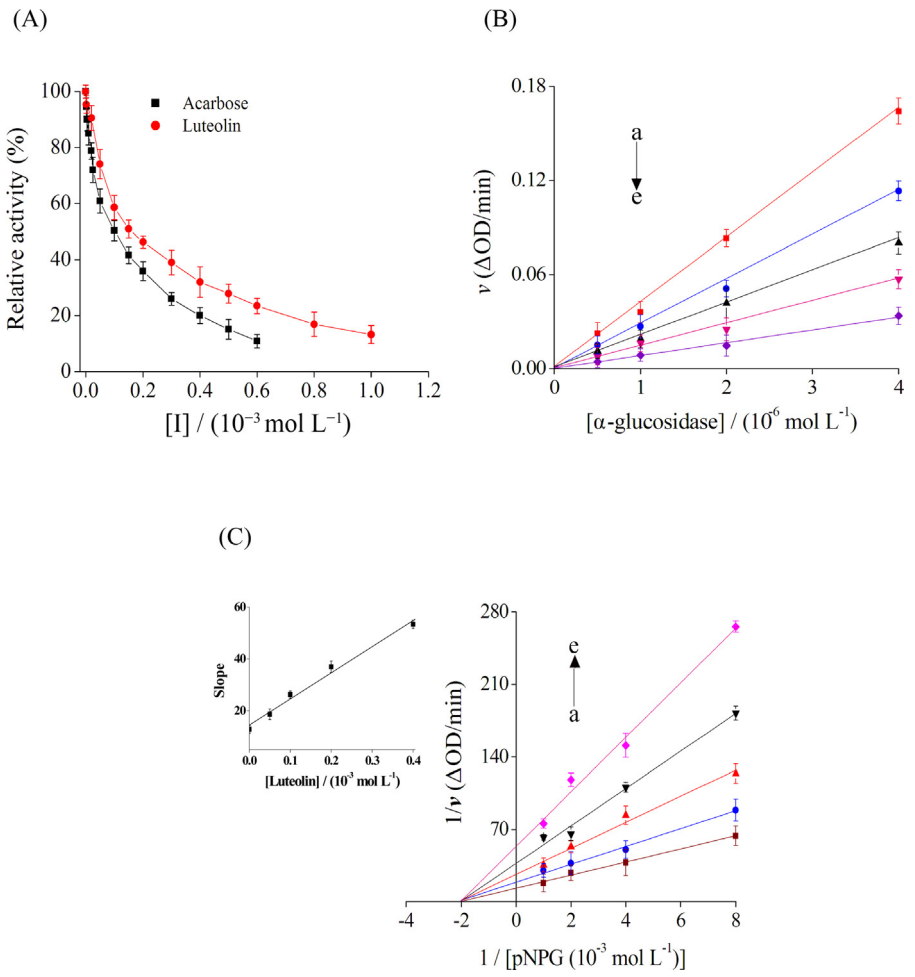


Fig. 2. Inhibitory effect of luteolin on α-glucosidase (pH 7.0, $T = 310$ K); (A) c(α-glucosidase) = 1.0×10^{-6} mol L^{-1}, and c(pNPG) = 5.0×10^{-4} mol L^{-1}. (B) Plots of v vs. [α-glucosidase]. c(Luteolin) = 0, 0.5, 1.0, 2.0 and 4.0×10^{-4} mol L^{-1} for curves a → e, respectively. (C) Lineweaver–Burk plots. c(α-Glucosidase) = 1.0×10^{-6} mol L^{-1}, and c(luteolin) = 0, 0.5, 1.0, 2.0 and 4.0×10^{-4} mol L^{-1} for curves a → e, respectively; The secondary plot of slope vs. [luteolin] was in the inset.

was enclosed in a grid having 0.375 Å spacing and inhibitor was allowed to move within the whole region to achieve the lowest energy conformation. Different energy terms including intermol energy (vdm + hbond + desolv energy + electrostatic energy), internal energy, torsional energy and binding energy can be obtained from the output which was rendered with PyMol from AutoDock, and estimated $\Delta G_{binding}$ = intermol energy + torsional energy [34].

2.10. Statistical analysis

The experimental results including α-glucosidase activity assay and inhibitory kinetic analysis were expressed as mean value ± standard deviation ($n = 3$), and data were analyzed using the SAS statistical package (version 8.1, SAS Institute, Cary, NC, USA).

3. Results and discussion

3.1. Effect of luteolin on α-glucosidase activity

It was found that luteolin inhibited α-glucosidase activity in a concentration dependent-manner (Fig. 2A). With increasing concentration of luteolin, the enzymatic activity decreased remarkably. The concentrations of luteolin and acarbose (positive control, structure shown in Fig. 1B) resulting in 50% activity lost (IC_{50}) were estimated to be $(1.72 \pm 0.05) \times 10^{-4}$ and $(1.08 \pm 0.04) \times 10^{-4}$ mol L^{-1}, respectively. The results indicated that luteolin has significant α-glucosidase inhibitory activity compared with acarbose.

The plots of the v vs. [α-glucosidase] at different inhibitor concentrations were constructed to confirm the reversibility of luteolin-mediated inhibition (Fig. 2B). As shown in Fig. 2B, the straight lines all passed through the origin, and the slope was decreased with increasing concentrations of the inhibitor. These results suggested that the inhibition of luteolin on α-glucosidase activity was reversible [35].

3.2. Type of inhibition

Lineweaver–Burk plots based on equation 1 were constructed and displayed in Fig. 2C. The value of horizontal axis intercept ($-1/K_m$) remained unchanged and the value of vertical axis intercept ($1/v_{max}$) increased with the increase of the concentrations of luteolin, indicating that luteolin was a noncompetitive inhibitor. The secondary plot of slope vs. [luteolin] (Eq. (2)) was linearly fitted (shown in the inset), indicating that luteolin had a single inhibition site on α-glucosidase, and the K_i value was calculated to be $(1.40 \pm 0.02) \times 10^{-4}$ mol L^{-1}.

3.3. Kinetics of α-glucosidase activity

The time-courses measurements were employed to determine the kinetics and rate constants. As shown in Fig. 3A, α-glucosidase

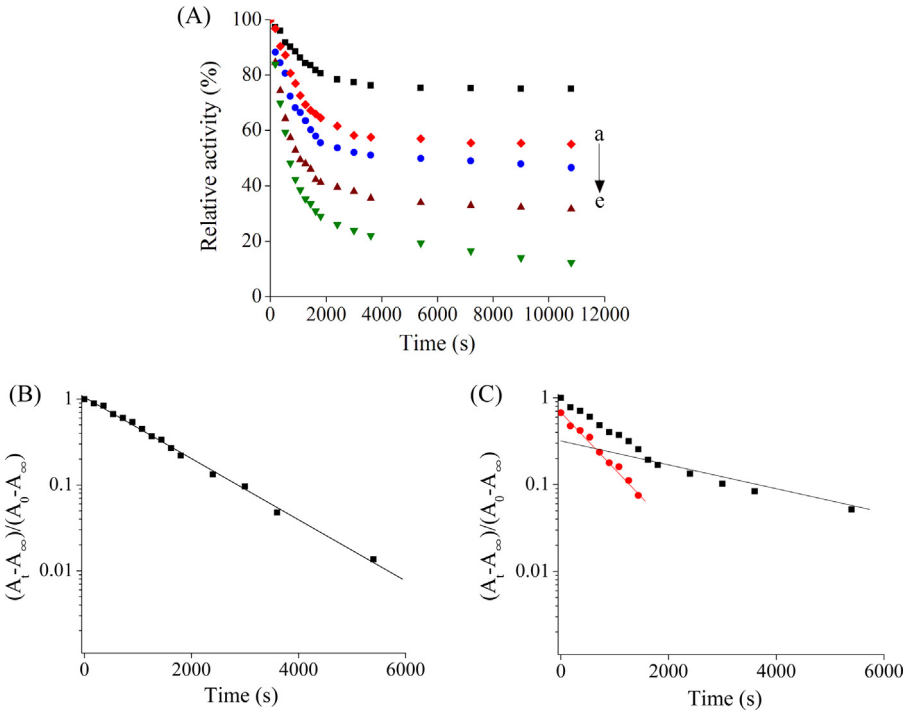


Fig. 3. (A) Kinetic time-courses for relative activity of α-glucosidase in the presence of luteolin (pH 7.0, T = 310 K) at 0.5, 1.0, 2.0, 4.0 and 10.0 × 10^{-4} mol L^{-1} for curves a → e, respectively. Aliquots were collected at the indicated time intervals. respectively. Semi-logarithmic plot analysis. Experimental points (■) were obtained in the presence of (B) 5.0 × 10^{-5} and (C) 2.0 × 10^{-4} mol L^{-1} luteolin, and points (●) were obtained by subtracting the contribution of the slow phase (black line) from the data in the curve. Slopes of the curves (red or black line) indicate the rate constants for fast and slow inactivation phases. The final concentrations of α-glucosidase and pNPG were 1.0 × 10^{-6} and 5.0 × 10^{-4} mol L^{-1}, respectively. (For interpretation of the references to color in this text, the reader is referred to the web version of the article.)

Table 1
Inactivation rate constants for α-glucosidase induced by luteolin.

Luteolin (×10^{-4} mol L^{-1})	Inactivation rate constants (×10^{-4} s^{-1})[a]			Transition free-energy change (kJ mol^{-1} s^{-1})[b]
	k_1	k_2	k	
0.5			2.49	21.39
1.0			2.55	21.33
2.0	4.86	0.32		19.66
4.0	5.17	0.33		19.51
10.0	5.57	0.33		19.31

[a] Data were calculated as shown in Fig. 3; k_1 and k_2 are the first-order rate constants for the fast and slow phases, respectively; k is the monophasic rate constant.
[b] Transition free-energy change per s. is $\Delta\Delta G^\circ = -RT\ln k$, where k is a time constant for the major phase of the inactivation reaction.

activity was decreased rapidly with increasing time from 0 to 2000 s at the various concentrations of luteolin, then followed by approximate stabilization with the further time increase. Kinetic analysis using semi-logarithmic plots [36] revealed monophasic (Fig. 3B) and biphasic inactivation (Fig. 3C) with fast (k_1) and slow (k_2) aspects, where a monophasic process developed into a biphasic process with increase in concentration of luteolin, indicating that α-glucosidase passed transiently through intermediates until it reached a inactive state. All processes allowed first-order kinetics and the inactivation rate constants are calculated and summarized in Table 1, It was also observed that the inactivation occurred as a result of changes in the transition free-energy energy ($\Delta\Delta G^\circ$), which decreased in a luteolin concentration-dependent manner.

3.4. Fluorescence quenching studies of α-glucosidase by luteolin

The significant inhibitory activity of luteolin on α-glucosidase suggested that luteolin may directly bind to the enzyme. To further explore the interaction between luteolin and α-glucosidase, fluorescence measurements were performed because it can give some information about the binding, such as binding mechanism, binding constants and binding sites.

The fluorescence quenching spectra of α-glucosidase by luteolin are shown in Fig. 4A. α-Glucosidase exhibited a strong fluorescence emission peak at 344 nm after being excited at a wavelength of 280 nm, while luteolin did not show intrinsic fluorescence under the same experimental conditions. With increasing amounts of luteolin to α-glucosidase solution, the fluorescence intensity of α-glucosidase at 344 nm decreased markedly (quenching ratio was approximately 84%, when the concentration of luteolin reached 3.54 × 10^{-4} mol L^{-1}), indicating that luteolin interacted with α-glucosidase, and quenched its intrinsic fluorescence.

The Stern–Volmer plots for the quenching of α-glucosidase by luteolin at three different temperatures (298, 304 and 310 K) are presented in Fig. 4B. Based on Eq. (4), the values of K_{SV} were obtained and listed in Table 2. The results showed that the K_{SV} values for the interaction between luteolin and α-glucosidase decreased with increasing temperature, and the corresponding K_q values were 10^{12} L mol^{-1} s^{-1}, which were two orders of magnitude greater than the limiting diffusion rate constant of the biomolecule (2 × 10^{10} L mol^{-1} s^{-1} [29]). Consequently, it could be concluded that the fluorescence quenching process was mainly controlled by a static quenching mechanism resulting from the formation of luteolin–α-glucosidase complex [29].

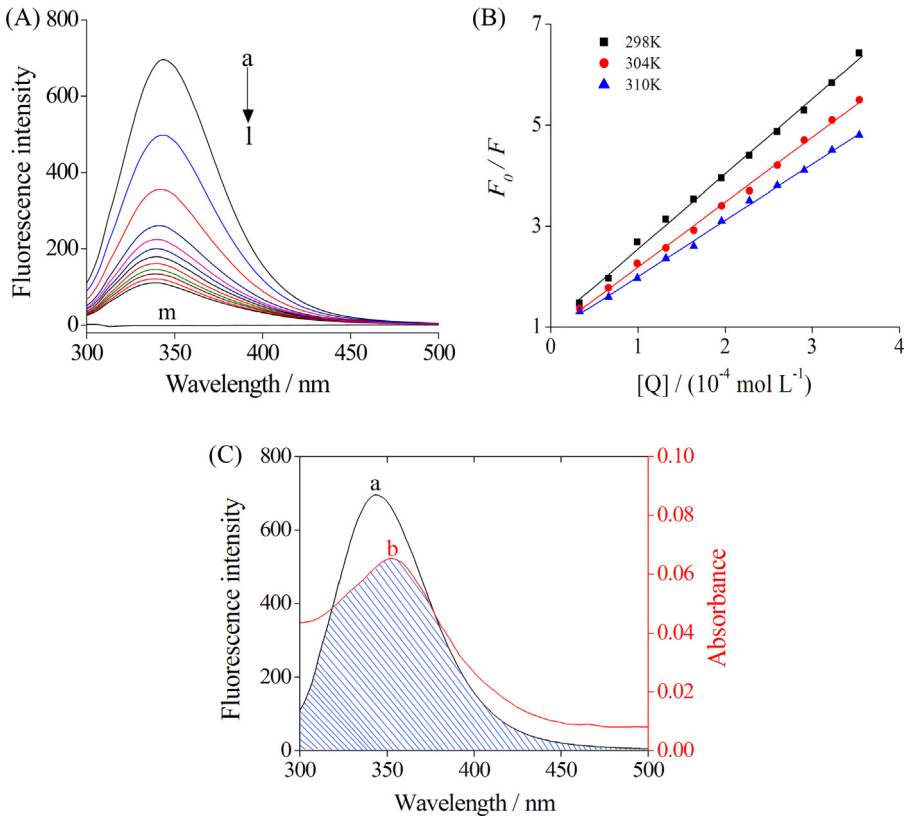


Fig. 4. (A) Fluorescence spectra of α-glucosidase in the presence of luteolin at different concentrations (pH 7.0, $T = 298$ K, $\lambda_{ex} = 280$ nm, $\lambda_{em} = 344$ nm). c(α-Glucosidase) = 2.0×10^{-6} mol L^{-1}, and c(luteolin) = 0, 0.33, 0.66, 0.99, 1.32, 1.64, 1.96, 2.28, 2.60, 2.91, 3.23 and 3.54×10^{-4} mol L^{-1} for curves a → l, respectively; Curve m shows the emission spectrum of luteolin only, c(luteolin) = 3.3×10^{-5} mol L^{-1}. (B) The Stern–Volmer plots for the fluorescence quenching at different temperatures. (C) Spectral overlaps of the fluorescence spectrum of α-glucosidase (a) with the absorption spectrum of luteolin (b). c(α-Glucosidase) = c(luteolin) = 2.0×10^{-6} mol L^{-1}.

Table 2
The quenching constants (K_{SV}), binding constants (K), number of binding sites (n) and relative thermodynamic parameters for the interaction between luteolin and α-glucosidase at different temperatures.

T (K)	K_{SV} ($\times 10^4$ L mol^{-1})	R^a	K ($\times 10^4$ L mol^{-1})	n	R^b	ΔH° (kJ mol^{-1})	ΔG° (kJ mol^{-1})	ΔS° (J mol^{-1} K^{-1})
298	1.49 ± 0.02	0.9985	1.55 ± 0.02	1.00 ± 0.02	0.9978		−23.91 ± 0.02	
304	1.28 ± 0.02	0.9988	1.78 ± 0.01	1.04 ± 0.01	0.9992	17.58 ± 0.1	−24.75 ± 0.02	139.23 ± 0.04
310	1.11 ± 0.01	0.9991	2.04 ± 0.01	1.08 ± 0.01	0.9992		−25.58 ± 0.02	

[a] R is the correlation coefficient for the K_{SV} values.
[b] R is the correlation coefficient for the K values.

In addition, the values of binding constant (K) and the number of binding sites (n) for luteolin–α-glucosidase complex at different temperatures were calculated on the basis of Eq. (5). The increasing trend of K with rising temperature was observed (Table 2), indicating that the capacity of luteolin binding to α-glucosidase was enhanced and the binding was an endothermic and entropy-driven reaction [23]. The values of n were approximately equal to 1, suggesting the existence of just a single binding site on α-glucosidase for luteolin, which convincingly supported the conclusion obtained from Lineweaver–Burk plot analysis.

3.5. Thermodynamic analysis and the nature of the binding forces

Generally, four types of forces exist in small ligands binding to biomolecules, viz., hydrogen bonds, van der Waals force, hydrophobic interaction and electrostatic forces. Table 2 lists the thermodynamic parameters for the interaction between luteolin and α-glucosidase. The negative signal for ΔG° indicated that the luteolin–α-glucosidase interaction was spontaneous. The values of ΔH° and ΔS° were 17.58 ± 0.1 kJ mol^{-1} and 139.23 ± 0.04 J mol^{-1} K^{-1}, respectively, suggesting that hydrophobic interactions played a predominant role in the interaction process [37]. However, hydrogen binding should not be ruled out because the experimental system was in aqueous solution and there are many hydroxyl groups on the molecules of flavonoid and protein [9].

3.6. Energy transfer between α-glucosidase and luteolin

The overlap of the UV–vis absorption spectrum of luteolin with the fluorescence emission spectrum of α-glucosidase is shown in Fig. 4C. According to the methods [9], the distance (r) between luteolin and α-glucosidase was determined to be 4.56 nm. Obviously, it was less than 8 nm, and $0.5R_0 < r < 1.5R_0$, implying that the energy transfer from α-glucosidase to luteolin occurred with high probability [38].

3.7. Change in conformation of α-glucosidase induced by luteolin

The shift in the maximum emission wavelength reflects the change of the polarity microenvironment around the tyrosine and tryptophan residues when the wavelength interval ($\Delta\lambda$) was set at

15 and 60 nm, respectively [39,40]. As shown in Fig. 5A, an obvious red shift (from 283 to 287 nm) of maximum emission wavelength of tyrosine residues was observed, while the fluorescence band of tryptophan residues displayed a slight red-shift from 284 to 286 nm (Fig. 5B). These results indicated that the conformation of α-glucosidase was somewhat changed and the polarity around the tyrosine and tryptophan residues increased and the hydrophobicity decreased during the binding process [41].

For confirming the conformational changes of α-glucosidase induced by luteolin, the CD spectra of α-glucosidase were measured with various amounts of luteolin (Fig. 5C). It was found that the spectra of α-glucosidase presented two negative bands at 209 and 222 nm, which are characteristic of α-helix of protein [42,43]. With the addition of luteolin, the CD intensity of α-glucosidase at 209 and 222 nm decreased (shifting to zero levels) without any significant shift of the peaks, which indicated partial changes in the protein secondary structure with the decrease in α-helical content. The contents of different secondary structures of α-glucosidase were calculated by the online SELCON3 program. As shown in Table 3, a decreasing tendency of α-helical content and an increasing tendency of β-sheet, β-turn and random coil structure contents were observed with increasing concentration of luteolin, suggesting rearrangement and conformational changes of α-glucosidase in the presence of luteolin [44]. Luteolin-induced secondary structure changes were similar to other α-glucosidase inhibitors. For example, curcuminoids analogs [42] and hydroxycoumarin derivatives [43] were reported to reduce the α-helical content in the enzyme molecule. These findings supported the proposition that the inhibitors induced the rearrangement of secondary structure and influenced the folding organization of α-glucosidase [43]. Hence, it might be part of the mechanism of the enzymatic inhibition required to prevent the hydration of the substrate binding site and induce the cleft closure to avoid entrance of the substrate [3].

Further evidence for the structural changes of α-glucosidase was obtained from FT-IR measurements. IR spectra of proteins exhibited a number of amide bands, which represent different vibrations of the peptide moiety. Among these amide bands of protein, amide I and amide II bands of IR spectra, which severally occur in the region 1700–1600 cm^{-1} (mainly C=O stretch) and 1600–1500 cm^{-1} (C—N stretch coupled with N—H bending mode), have been extensively used in the characterization of chemical composition and conformational studies of proteins [45]. As shown in Fig. 6A, the peak positions of amide I (1659 cm^{-1}) and amide II (1548 cm^{-1}) bands of free α-glucosidase were shifted to 1653 cm^{-1} and 1541 cm^{-1} upon the addition of luteolin. The changes of these peak positions suggested that luteolin interacted with the C=O and C—N groups in the protein structural subunits, which induced the rearrangement of the polypeptide carbonyl hydrogen bonding pattern [46].

In general, amide I band is more widely used in the studies of protein secondary structures because it is more sensitive to the change of protein secondary structures than amide II band [47]. According to the well-established assignment criterion, the spectral ranges of 1615–1637 cm^{-1}, 1638–1648 cm^{-1}, 1649–1660 cm^{-1}, 1660–1680 cm^{-1} and 1680–1692 cm^{-1} are generally assigned to β-sheet, random coil, α-helix, β-turn and β-antiparallel, respectively [45,48]. The percentages of each secondary structure of α-glucosidase were calculated from the integrated areas of component bands in amide I band. The free α-glucosidase contained 37.9% α-helix, 23.3% random coil, 14.6% β-turn, 9.8% β-sheet and 14.4% β-antiparallel (Fig. 6B). When the added concentration of luteolin was 4.0×10^{-4} mol L^{-1}, the content of α-helix and β-antiparallel

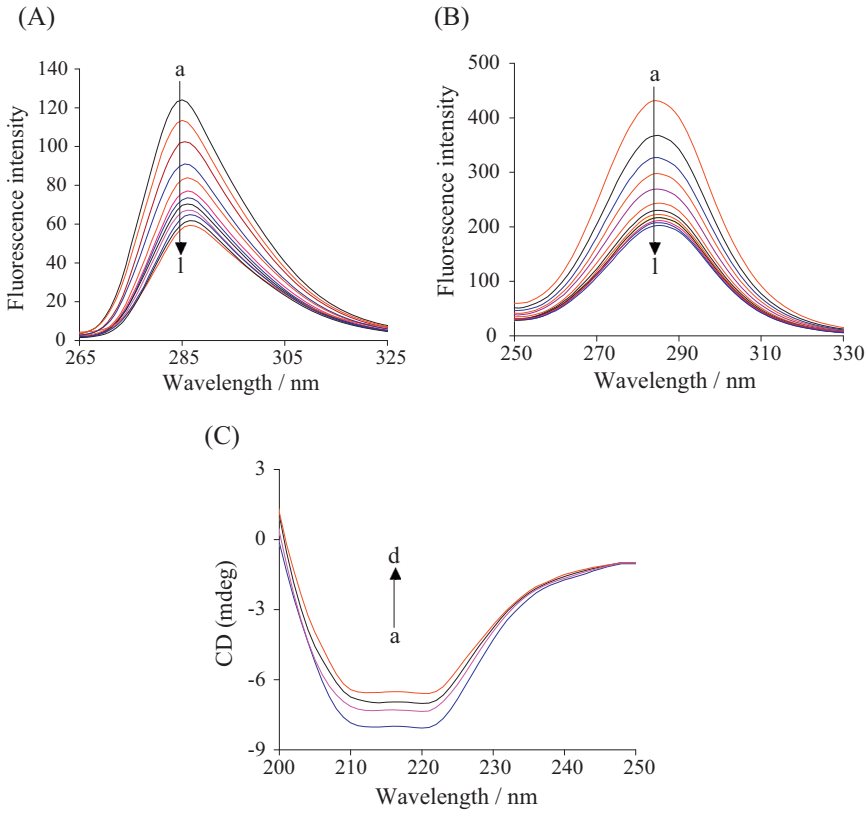


Fig. 5. The synchronous fluorescence spectra of α-glucosidase in the absence and presence of luteolin (A) Δλ = 15 nm, (B) Δλ = 60 nm (pH 7.0, T = 298 K). c(α-Glucosidase) = 2.0×10^{-6} mol L^{-1}, and c(luteolin) = 0, 0.33, 0.66, 0.99, 1.32, 1.64, 1.96, 2.28, 2.60, 2.91, 3.23 and 3.54×10^{-4} mol L^{-1} for curves a → l, respectively; (C) CD spectra of the α-glucosidase–luteolin system in the presence of increasing amounts of luteolin at pH 7.0, room temperature. c(α-Glucosidase) = 2.0×10^{-6} mol L^{-1}, and the concentrations of luteolin were 0, 1.0, 2.0 and 4.0×10^{-4} mol L^{-1} for curves a → d, respectively.

Table 3
The contents (CD spectra) of different secondary structures of free α-glucosidase and luteolin–α-glucosidase systems at pH 7.0.

Concentration luteolin ($\times 10^{-4}$ mol L^{-1})	α-Helix (%)	β-Sheet (%)	β-Turn (%)	Random coil (%)
0	37.9	23.0	14.3	24.8
1.0	34.0	25.5	15.2	25.3
2.0	31.6	26.2	16.7	25.5
4.0	30.0	27.0	17.3	25.7

decreased from 37.9% to 30.2% and from 14.4% to 8.5%, respectively, while the contents of random coil, β-turn and β-sheet increased from 23.3% to 26.1%, from 14.6% to 16.9% and from 9.8% to 18.3%, respectively (Fig. 6C). These results are consistent with that obtained from CD analysis.

3.8. Computational docking of luteolin on α-glucosidase

The sequence alignment between α-glucosidase (gi number 411229) from baker's yeast and isomaltase (PDB ID: 3A4A) from *Saccharomyces cerevisiae* is shown in Fig. 7A. Their sequence identity and similarity were 73% and 85%, respectively. The high sequence homology suggested that a high-quality 3D structure of α-glucosidase can be expected in the homology modeling [49].

A cluster analysis was performed using a rmsd tolerance of 2.0 Å (Fig. 7B). A total of 29 multimember conformational clusters were obtained from 100 docking runs. The most energetically favorable cluster possessed the lowest energy and the most frequent locus (red histogram) was used for binding orientation analysis. As shown in Table 4, the calculated binding energy (−5.33 kcal/mol) was the same as the predicted $\Delta G_{binding}$ (−5.33 kcal/mol), indicating that the docking result was credible. The binding energy was a little larger than ΔG° (−5.72 kcal/mol) obtained from the thermodynamic analysis at 298 K. As the molecular docking was conducted under simulation of vacuum environment, the difference of the binding energy between docking results and experiment results could be attributed to the lack of desolvation energy.

The docking result (Fig. 8A) showed that the binding site on α-glucosidase for luteolin was not the same as the binding site for acarbose (blue region, active site) [50]. Indeed, the docking analysis (Fig. 8B) predicted that acarbose, as a competitive inhibitor [32], was surrounded by residues Glu277, His351 and Asp352, which are part of the catalytic residues of isomaltase [51], and these residues are believed to play critical roles in the catalytic mechanism as the corresponding residues of Glu276, His348 and Asp349 in α-glucosidase [52]. It was also observed that the binding site for luteolin was close to the active site pocket (Fig. 8B), and luteolin was surrounded by the residues, namely, Phe303, Ser304, His305, Val308, Gly309, Thr310, Ser311 and Pro312. Moreover, one hydroxyl group on A ring could form hydrogen bonds (green line) with residue Thr310, indicating that hydrogen bond was another main role in the binding of luteolin with α-glucosidase. In addition, the 3′,4′-dihydroxyphenyl moiety at B ring of luteolin was inserted

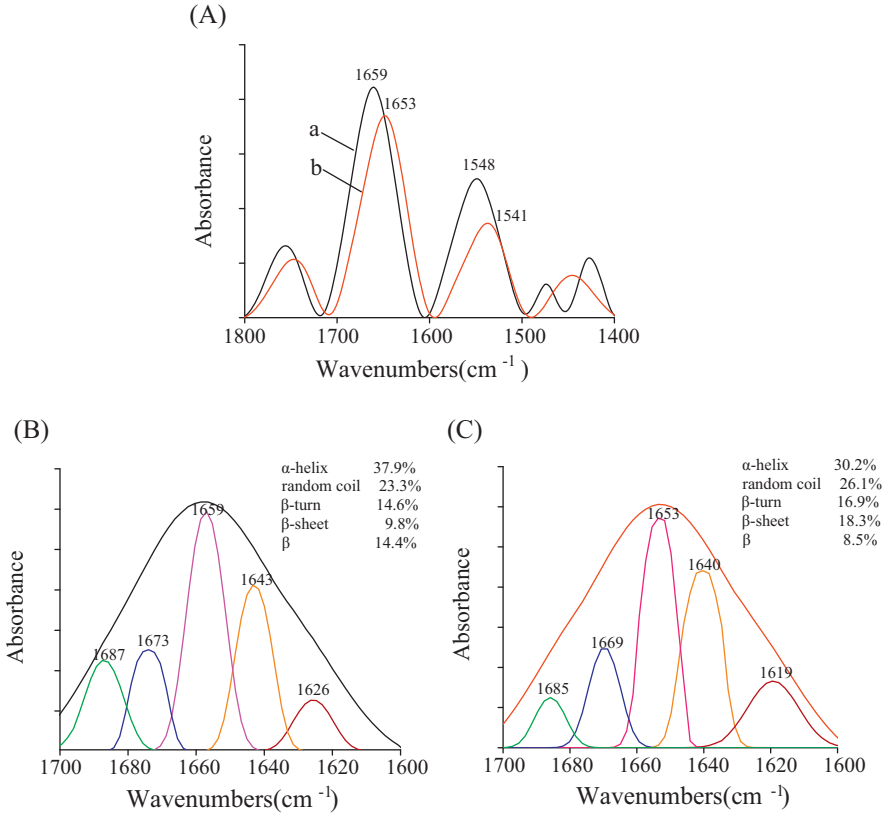
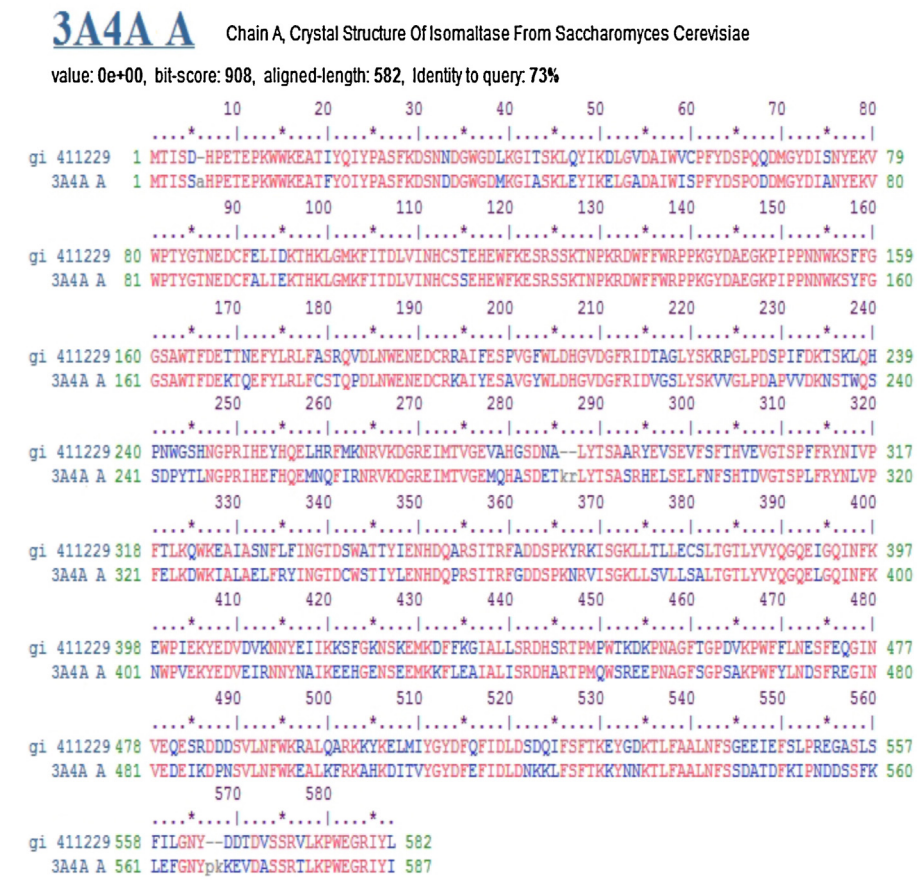


Fig. 6. (A) The FT-IR spectra of free α-glucosidase (a) and difference spectra [(luteolin–α-glucosidase solution) – luteolin solution] (b) at pH 7.0 and room temperature in the region of 1800–1400 cm^{-1}. c(α-Glucosidase) = 2.0 × 10^{-6} mol L^{-1}, and c(luteolin) = 4.0 × 10^{-4} mol L^{-1}. The curve-fitted amide I region of free α-glucosidase (B) and its luteolin complex (C).

(A)



(B)

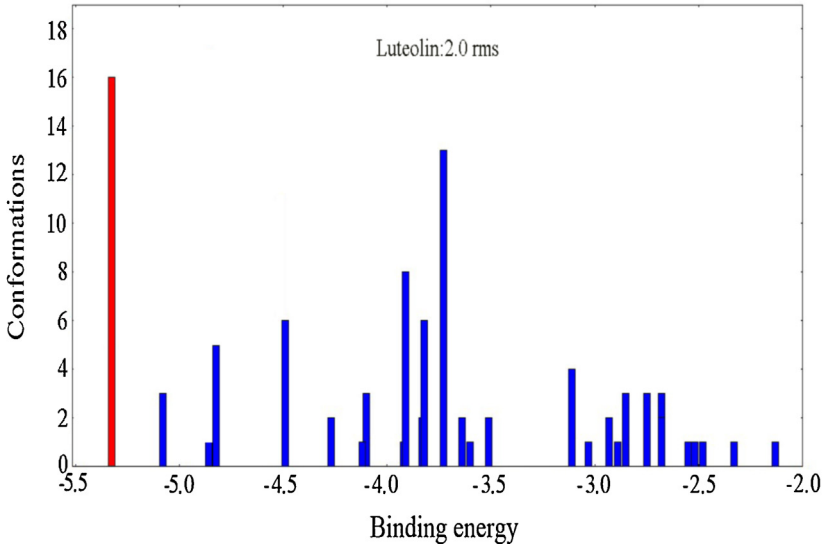


Fig. 7. (A) Sequence alignment between α-glucosidase and isomaltase (3A4A). The identity between the corresponding residues was indicated in a red color. (B) Cluster analysis of the AutoDock docking runs of luteolin with α-glucosidase. (For interpretation of the references to color in this text, the reader is referred to the web version of the article.)

into the hydrophobic region to interact with residues Phe303, Ser311, Pro312 and His351 which were closed to the pocket of active site, and might contribute to inducing the cleft closure to avoid entrance of the substrate. The molecular docking provided supportive data for luteolin-induced inhibition by predicting the binding site on α-glucosidase based on the lowest energy principle. The docking results indicated that luteolin actually interacted with the amino acid residues located in the region close to the active

Table 4
The binding free energies of luteolin and α-glucosidase according to docking result.

Complex	Intermol energy (kcal/mol)	Internal energy (kcal/mol)	Torsional energy (kcal/mol)	Binding energy (kcal/mol)	Estimated $\Delta G_{binding}$ (kcal/mol)
Luteolin–enzyme	−5.63	−0.27	0.30	−5.33	−5.33

Estimated $\Delta G_{binding}$ is the sum of intermolecular energy and torsional free energy.

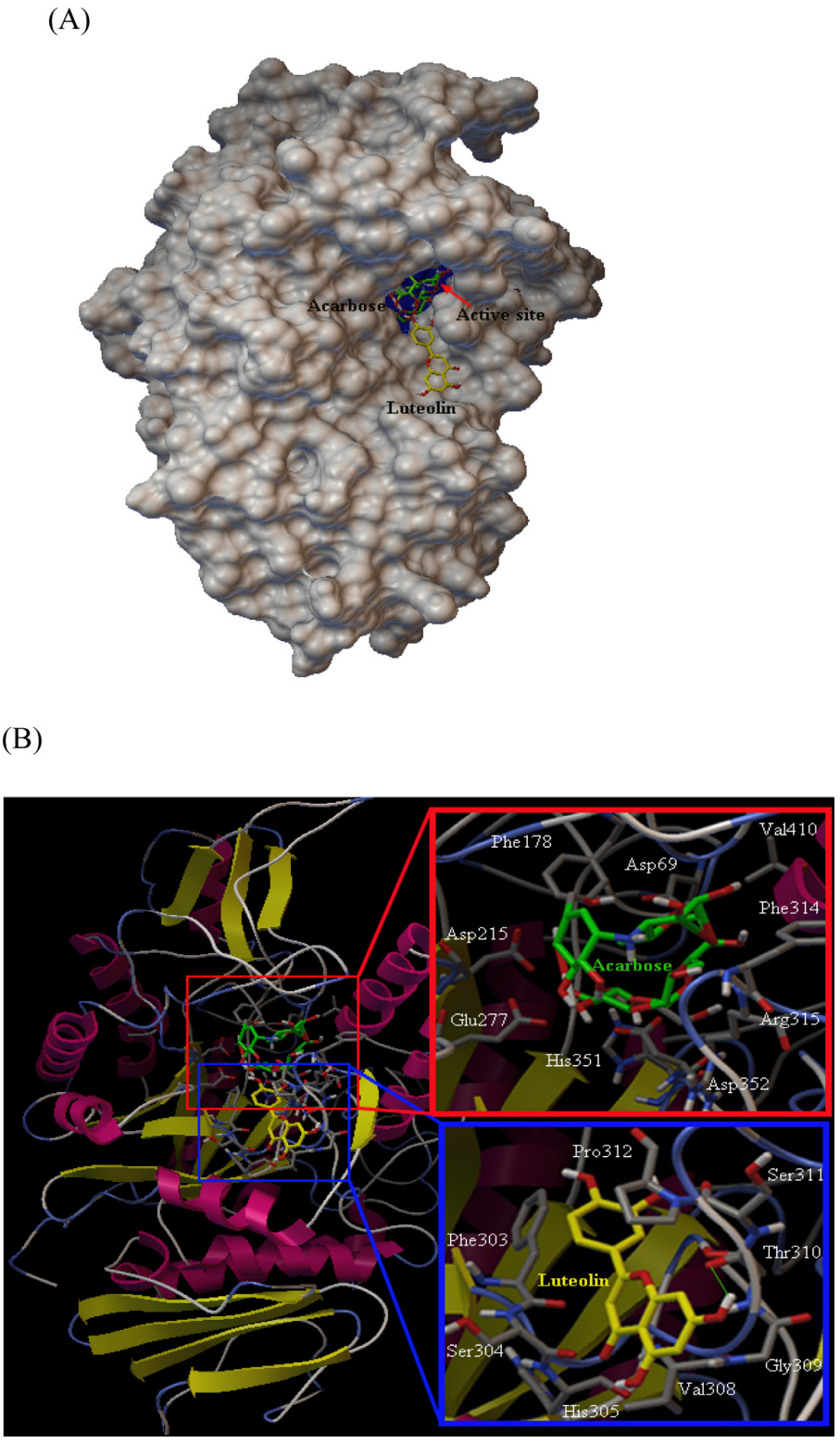


Fig. 8. (A) Predicted binding mode of luteolin docked with α-glucosidase on molecular surface. (B) The interaction between luteolin and α-glucosidase. The residues of α-glucosidase and the luteolin structure were represented using stick structures. The dashed lines (green) represent hydrogen-bonding interactions. (For interpretation of the references to color in this text, the reader is referred to the web version of the article.)

site of α-glucosidase, which confirmed luteolin as a noncompetitive inhibitor with a single high-affinity binding site on α-glucosidase.

4. Conclusions

Luteolin showed a significant inhibitory activity on α-glucosidase, and the half maximal inhibitory concentration (IC_{50}) of luteolin was $(1.72 \pm 0.05) \times 10^{-4}$ mol L^{-1}. The inhibition type of luteolin was a noncompetitive inhibition with K_i value of $(1.40 \pm 0.02) \times 10^{-4}$ mol L^{-1}. The fluorescence quenching of α-glucosidase was considered as a static procedure resulting from the formation of the luteolin–α-glucosidase complex. The thermodynamic parameters, ΔH° and ΔS°, were calculated to be 17.58 ± 0.1 kJ mol^{-1} and 139.23 ± 0.04 J mol^{-1} K^{-1}, respectively, suggesting that hydrophobic interactions played a predominant role in the binding of luteolin to α-glucosidase. The values of n revealed the presence of a single class of binding site on α-glucosidase for luteolin, and the binding distance between α-glucosidase and luteolin was 4.56 nm based on Förster theory, which suggested that the energy transfer from α-glucosidase to luteolin occurred with high possibility. The results of synchronous fluorescence, CD and FT-IR spectra indicated that binding of luteolin to α-glucosidase induced rearrangement and conformational changes of the enzyme. The molecular docking studies showed that the binding site on α-glucosidase for luteolin was not the same as acarbose (a competitive inhibitor), while luteolin had a high affinity close to the active site pocket of the enzyme. Our study has provided useful information of the effects of luteolin on α-glucosidase and the characteristics of binding of luteolin with α-glucosidase. A combination of inhibition kinetics, interaction mechanism and molecular modeling may facilitate the evaluation of inhibitory mechanism. The results presented here indicated that luteolin could be a promising α-glucosidase inhibitor for further animal studies or clinical trials. We believe that the ingestion of luteolin-rich foods such as celery, green pepper and carrot may be useful to the treatment of type 2 diabetes mellitus.

Acknowledgements

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